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Application self-reporting for application status monitoring in enterprise networks

Abstract

A computing device within an enterprise network is described, the computing device configured to execute one or more applications and one or more status modules embedded within the one or more applications. A first status module embedded within a first application running on the computing device is configured to periodically determine an operational status of the first application based on at least one value of at least one metric of the first application for each periodic interval. The at least one metric may be selected from a predefined list of metrics as being indicative of the operational status of the first application. The first status module is configured to report an identifier of the computing device on which the first application is running and a status indicator representative of the operational status of the first application at a given time to an application performance tool within the enterprise network.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to computer networks, and more specifically, to monitoring of applications running on computing devices within a network.

BACKGROUND

(2) Enterprise networks, especially large enterprise networks, require significant efforts to maintain and administer. Such networks typically host many different types of applications and systems, each of which may evolve through continual updates, modifications, and bug fixes. In addition, such networks may host multiple different instances of the same application or system (e.g., active-standby instances and/or active-active instances) on the mainframe and/or at multiple geographically-disparate data centers. The continually changing nature of a typical enterprise network and the bandwidth cost required to continually, or even frequently, scan the applications and systems hosted on the enterprise network makes planning and responding to adverse events (e.g., application or data center failures) difficult.

SUMMARY

(3) This disclosure describes an enterprise network including a centralized application performance tool and a plurality of computing devices (e.g., data center servers and/or mainframe computers) configured to execute one or more status modules embedded within one or more applications. A status module embedded within an application is configured to periodically determine an operational status of the application based on at least one value of at least one metric of the application for each periodic interval, and self-report a status indicator representative of the operational status of the application at a given time to the centralized application performance tool. For example, the status module may be configured to determine whether the application is an active instance of a distributed application within the enterprise network based on the at least one metric. In other examples, the status module may be configured to determine whether the application's performance is healthy and/or to compute a composite health score for the application based on one or more metrics. In one example, the status module may self-report the status indicator by recording the status indicator, and in some cases the value of the metric, as a timestamped entry in a dedicated status log file for the application. In this example, the status log file may then be read, packaged for transmission, stored in a central database, or otherwise reported to the centralized application performance tool.

(4) The centralized application performance tool receives the status indicators from one or more of the plurality of applications. In one example, the application performance tool may read or otherwise receive the status log files including the status indicators and the corresponding metric values from the applications. In some examples, the application performance tool may receive the status log files from the applications via a third-party monitoring tool included in the enterprise network. According to the disclosed techniques, the application performance tool may perform comparisons or other analyses of the status indicators and/or the metric values across multiple instances of the same application in order to verify which instance is the active or primary instance and/or which instances are the healthiest. The application performance tool may generate a user interface to display locations (e.g., data center and/or mainframe) and statuses of the plurality of applications running in the enterprise network. The user interface may enable a user, e.g., an administrator of the enterprise network, to view the application statuses from a data center perspective of what applications are running on a certain data center, and/or from an application perspective of where each instance of a given application is running.

(5) The techniques described in this disclosure provide one or more technical advantages and practical applications. For example, available tools for application discovery, including third-party monitoring tools, typically rely on scanning the entire network environment, which adds overhead on application servers and requires network bandwidth. In addition, such scanning tools may only be able to discover which applications are running in the network environment, and not where the applications are running (e.g., data center servers and/or mainframe computers) and/or where the active or primary instance of a distributed application is running. The disclosed status modules comprise lightweight tools (e.g., code modules) that may replace or supplement third-party network monitoring tools to determine location and status of applications running within an enterprise network without the use of bandwidth- and resource-intensive scanning.
(6) In one example, this disclosure is directed to a computing device within an enterprise network, the computing device comprising a storage device configured to store one or more applications and one or more status modules embedded within the one or more applications; and processing circuitry having access to the storage device and configured to execute instructions associated with the one or more status modules embedded within the one or more applications. To execute instructions associated with a first status module embedded within a first application, the processing circuitry is further configured to periodically determine an operational status of the first application based on at least one value of at least one metric of the first application for each periodic interval, wherein the at least one metric is selected as being indicative of the operational status of the first application; and report an identifier of the computing device on which the first application is running and at least a status indicator representative of the operational status of the first application at a given time to an application performance tool within the enterprise network.
(7) In another example, this disclosure is directed to a method comprising periodically determining, by a first status module embedded within a first application of one or more status modules embedded within one or more applications running on a computing device of a plurality of computing devices within an enterprise network, an operational status of the first application based on at least one value of at least one metric of the first application for each periodic interval, wherein the at least one metric is selected as being indicative of the operational status of the first application; and reporting, by the first status module, an identifier of the computing device on which the first application is running and at least a status indicator representative of the operational status of the first application at a given time to an application performance tool within the enterprise network.
(8) In a further example, this disclosure is directed to a network system comprising an application performance tool; and a plurality of computing devices, each computing device configured to execute one or more status modules embedded within one or more applications. Each status module embedded within a respective application running on the respective computing device is configured to periodically determine an operational status of the respective application based on at least one value of at least one metric of the respective application for each periodic interval, wherein the at least one metric is selected as being indicative of the operational status of the respective application, and report an identifier of the respective computing device on which the respective application is running and at least a status indicator representative of the operational status of the respective application at a given time to the application performance tool. The application performance tool is configured to analyze status indicators received from the status modules running on the plurality of computing devices to identify, for a particular application at the given time, a location of a primary instance of the particular application within the network system.
(9) The details of one or more examples of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram illustrating an example enterprise network including a centralized application performance tool and a plurality of status modules configured to self-report statuses of respective applications running within the enterprise network, in accordance with one or more aspects of the present disclosure.

(2) FIG. **2** is a block diagram illustrating an example server device executing applications having embedded status modules configured to self-report application statuses, in accordance with one or more aspects of the present disclosure.

(3) FIG. **3** is a block diagram illustrating an example status module configured to determine an operational status of a respective application based on at least one value of at least one metric and self-report a status indicator for the application, in accordance with one or more aspects of the present disclosure.

(4) FIG. **4** is a block diagram illustrating an example computing device executing an application performance tool configured to process application statuses received from a plurality of applications, in accordance with one or more aspects of the present disclosure.

(5) FIG. **5** is a flow diagram illustrating operations performed by an example server executing one or more applications and embedded status modules, in accordance with one or more aspects of the present disclosure.

(6) FIG. **6** is a flow diagram illustrating operations performed by an example computing device executing an application performance tool, in accordance with one or more aspects of the present disclosure.

DETAILED DESCRIPTION

(7) FIG. **1** is a block diagram illustrating an example enterprise network **100** including a centralized application performance tool **110** and a plurality of status modules **106**A-**106**M configured to self-report statuses of respective applications **104**A-**104**M running within enterprise network **100**, in accordance with one or more aspects of the present disclosure. Enterprise network **100** may be a large-scale enterprise network used or administered by a large organization, such as a financial institution, bank, medical facility, or other type of large organization, which will commonly rely on significant computing resources. For some large organizations, the required computing resources are provided through multiple data centers deployed within the enterprise network.

(8) Accordingly, as illustrated in FIG. **1**, enterprise network **100** is generally characterized by multiple data centers **102**A-**102**K (collectively, “data centers **102**”) and at least one mainframe computer **112**. Data centers **102**, computing devices within each of data centers **102**, and/or mainframe computer **112** may host enterprise applications and provide a platform for execution of applications and services provided to users of user devices **116**A-**116**N (collectively, “user devices **116**”). User devices **116** may interact with mainframe **112** through network **114** and may interact with servers within data centers **102** through network **114** and respective data center networks of data centers **102**. For ease of illustration, only data center network **122**A of data center **102**A is illustrated in FIG. **1**. It should be understood that each of data centers **102**B-**102**K also include a data center network **122**B-**122**K to interconnect computing devices within the respective data center, for example data center network **122**B should be understood to be included within data center **102**B and data center network **122**K should be understood to be included within data center **102**K. One or more client devices **116** may be operated by users of enterprise network **100**, and may access functionality of enterprise network **100**, generally provided by data centers **102** and/or mainframe **112**. In some examples, each of data centers **102** may be located in a geographically-disparate location in order to provide high-speed services to user devices **116** in the same geographic location.

(9) Network **114** illustrated in FIG. **1** may include or represent any public or private communications network or other network. One or more client devices, server devices, or other devices may transmit and receive data, commands, control signals, and/or other information across such networks using any suitable communication techniques. In some examples, network **114** may be a separate network as illustrated in FIG. **1**, or one or more of such networks may be a subnetwork of another network. In other examples, two or more of such networks may be combined into a single network; further, one or more of such networks may be, or may be part of, the internet. Accordingly, one or more of the devices or systems illustrated in FIG. **1** may be in a remote location relative to one or more other illustrated devices or systems. Network **114** illustrated in FIG. **1** may include one or more network hubs, network switches, network routers, network links, satellite dishes, or any other network equipment. Such devices or components may be operatively inter-coupled, thereby providing for the exchange of information between computers, devices, or other components (e.g., between one or more user devices or systems and one or more server devices or systems).
(10) In accordance with the disclosed techniques, data center **102**A includes one or more computing devices, e.g., servers **108**A-**108**X (collectively, "servers **108**"), configured to execute applications **104**A-**104**M (collectively, "applications **104**") and status modules **106**A-**106**M (collectively, "status modules **106**") embedded within respective applications **104**. For ease of illustration, only the applications **104** and status modules **106** within server **108**A of data center **102**A are illustrated in FIG. **1**. It should be understood that each of servers **108**B-**108**X may also execute applications having embedded status modules. In addition, it should be understood that similar sets of computing devices executing applications having embedded status modules may be included within each of data centers **102**B-**102**K. Furthermore, it should be understood that mainframe computer **112** may also be configured to execute applications having embedded status modules similar to those illustrated in server **108**A of data center **102**A.
(11) Application performance tool **110** may receive interactions from console **126** and may perform functions in response to input received from console **126**. Application performance tool **110** also communicates with, and/or has access to, enterprise data store **128**. Enterprise data store **128** may represent any suitable data structure or storage medium for storing information related to enterprise network **100** or systems, devices, or applications included within enterprise network **100**. In some examples, enterprise data store **128** may represent a system of record associated with an enterprise network **100**, which may serve as an authoritative data source for at least some data pertaining to enterprise network **100**, or pertaining to the operations of the business, organization, or other entity that administers enterprise network **100**.
(12) In some examples, enterprise data store **128** may be updated and/or maintained by application performance tool **110**. The information stored in enterprise data store **128** may be searchable and/or categorized such that one or more modules of application performance tool **110** may provide an input requesting information from enterprise data store **128**, and in response to the input, receive information stored within enterprise data store **128**. Enterprise data store **128** may include one or more application information records and an application metric value store. In some examples, aspects of enterprise data store **128** may be included within application performance tool **110**. In other examples, some or all of enterprise data store **128** may be accessed through a separate system.
(13) In operation, application performance tool **110** receives information for an application, e.g., application **104**A, that is deployed, stored, and/or executing within one or more data centers **102** and/or mainframe **112**. For instance, for an enterprise that has multiple lines of business, each line of business may develop applications for execution on the shared enterprise network **100** (e.g., within one or more of data centers **102** and mainframe **112**) used by each line of business within the enterprise. Typically, each such line of business is responsible for maintaining certain information for any applications that are used by that line of business from sunrise of the application (initial development of the application) to sunset of the application (phasing out or

shutting down of redundant or obsolete business applications).

(14) For instance, each line of business may maintain, for each application, a list of resources needed by the application for proper performance (e.g., CPU, memory, and other resource requirements). Each line of business may also maintain a list of dependencies (e.g., data required, internal hardware, software, and/or databases required for operation) that the enterprise application relies upon to operate effectively and perform services on behalf of user devices **116**. In accordance with the techniques described in this disclosure, each line of business may further maintain guidelines for how the application is to be monitored on a periodic basis. For example, each line of business may maintain which metrics of the enterprise application are indicative of an operational status (e.g., active/inactive, healthy/unhealthy, or a composite health score) of the enterprise application and, in some cases, thresholds and/or ranges of metric values associated with certain operational statuses. Finally, each line of business may also maintain further information about the enterprise application, including how it is deployed, usage patterns (e.g., scheduled or typical operational periods of the enterprise application), and/or historical information. Maintenance of such information may involve creating and updating one or more application information records included within enterprise data store **128** for enterprise network **100** illustrated in FIG. **1**.

(15) Application performance tool **110** may receive information about applications from the line of business or other source through console **126**, or through another channel or system. As changes, modifications, or updates to enterprise applications are made, application performance tool **110** may receive further information about enterprise application **104**A. As the enterprise application evolves during its lifecycle, application performance tool **110** may use such further information to update enterprise data store **128**, and may include some or all of such information within the application information records. Accordingly, in some examples, a business or other entity may maintain enterprise data store **128** and keep within enterprise data store **128** up-to-date information about some or all of the many enterprise applications that may execute within data centers **102** and/or mainframe **112**.

(16) When enterprise application **104**A is deployed and in use, e.g., within server **108**A of data center **102**A, enterprise application **104**A is accessible to one or more of user devices **116** that may request that enterprise application **104**A perform services on their behalf. For instance, in one example, one or more of user devices **116** may interact with server **108**A executing enterprise application **104**A within data center **102**A. Enterprise application **104**A may receive one or more indications of input that it determines correspond to input from a user of a user device, e.g., user device **116**A. In response to the input, enterprise application **104**A causes server **108**A to perform operations and services on behalf of the user of client device **116**A. In addition, enterprise application **104**A may log events related to the performance of the operations and services in one or more application logs (not shown in FIG. **1**). In some examples, enterprise application **104**A may further compute certain metrics based on the logged events over time. For example, enterprise application **104**A may continuously or periodically compute metrics including service call volume, service call speed, and/or data write frequency based on activity in application logs, CPU throughput of server **108**A, and/or IO (input/output) speed of server **108**A.

(17) Large-scale enterprise networks, such as enterprise network **100** illustrated in FIG. **1**, typically provide computing resources required to host enterprise applications and provide functionality to user devices through multiple geographically-disparate data centers deployed in the enterprise network. In addition, such enterprise networks may host multiple different instances of the same application or system (e.g., active-standby instances and/or active-active instances) on a mainframe and/or at the multiple geographically-disparate data centers. Available tools for application discovery, e.g., third-party monitoring tool **124**, typically rely on scanning the entire network environment, which adds overhead on application servers and requires network bandwidth. In addition, such tools may only be able to discover which applications are running in the network environment, and not where the applications are running (e.g., data center servers and/or

mainframe computers) and/or where the active or primary instance of a distributed application is running. The continually changing nature of typical enterprise networks and the bandwidth cost required to continually, or even frequently, scan the applications and systems hosted on the enterprise network makes planning and responding to adverse events (e.g., application or data center failures) difficult.

(18) This disclosure describes status modules **106** as lightweight code modules embedded within respective enterprise applications **104** and configured to self-report statuses of the respective applications **104** running within enterprise network **100**. According to the disclosed techniques, a given status module, e.g., status module **106**A embedded within application **104**A, is configured to periodically determine an operational status of respective application **104**A based on at least one value of at least one metric of application **104**A for each periodic interval, and self-report a status indicator representative of the operational status of application **104**A at a given time to centralized application performance tool **110**. Status modules **106** may replace or supplement third-party network monitoring tools, e.g., third-party monitoring tool **124**, to determine location and status of applications **104** running within enterprise network **100** without the use of bandwidth- and resource-intensive scanning.

(19) Each of status modules **106** for a given platform, e.g., Java, Windows, etc., may be programmed or configured with a predefined list of metrics, e.g., 3 to 10 different metrics, based on which the status module is able to determine an operational status of applications for the given platform. Since applications, even of the same platform, may vary with respect to types of service calls performed, quantity and frequency of administrative functions, and data logging and storage, the one or more metrics used to determine the operational status of the application may be application-specific. As such, in order to initialize each of status modules **106** to determine the operational status of their respective applications **104**, application performance tool **110** may send a message to each of status modules **106** identifying the metric or metrics that are indicative of the operational status of their respective applications **104**. In some examples, the message to each status modules **106** may also identify thresholds and/or ranges of metric values associated with certain operational statuses. As noted above, each line of business may maintain metrics and thresholds and/or ranges for each of their applications that are indicative of an operational status and provide that information to application performance tool **110**, e.g., via console **124**, and/or store that information in application information records within data store **128**.

(20) Status module **106**A embedded within application **104**A is configured to periodically determine the operational status of application **104**A based on a value of a selected metric of application **104**A for each periodic interval. Status module **106** may select the metric from the predefined list of metrics based on the initialization message received from application performance tool **110** identifying the metric as being indicative of the operational status of application **104**A. For example, status module **106**A may determine whether application **104**A is an active instance of a distributed application within enterprise network **100** based on the selected metric. In other examples, status module **106**A may determine whether the performance of application **104**A is healthy based on the selected metric. In further examples, status module **106**A may compute a composite health score for application **104**A based on two or more selected metrics.

(21) Status module **106**A may self-report an identifier of server **108**A on which application **104**A is running and a status indicator representative of the operational status of application **104**A at a given time, either periodically or in response to a request. The status indicator may be a one-bit flag, a multi-bit indicator, or a binary representation of a numerical value. In some examples, status module **106**A may self-report a single status indicator of application **104**A representative of one of: activity or liveness, performance health, or composite health. In other examples, status module **106**A may self-report two or more status indicators. For example, status module **106**A may transmit the status indicator in a message to application performance tool **110**. The message may be a dedicated status reporting message generated by status module **106**A or may be an existing

reporting message of application **104**A onto which the status indicator may be added. As another example, status module **106**A may store the status indicator in data store **128** or another central database of enterprise network **100** that is accessible by application performance tool **110**. In a further example, status module **106**A may provide read access to the status indicator via an interface (e.g., an application programming interface (API)) of application **104**A for at least one of application performance tool **110** or third-party monitoring tool **124** within enterprise network **100**. In examples where third-party monitoring tool **124** reads the status indicator from application **104**A on data center server **108**A, third-party monitoring tool **124** may subsequently report the status indicator to application performance tool **110**.

(22) In one specific example, status module **106** may self-report the status indicator by recording the status indicator, and in some cases the value of the metric used to determine the operational status of application **104**A at each periodic interval, as a timestamped entry in a dedicated status log (not shown in FIG. **1**) for application **104**A. In this example, status module **106**A may self-report the status indicator for application **104**A at the given time by packaging the entries in the status log at the given time for transmission, storing the entries of the status log in data store **128**, providing read access to the status log file, or otherwise reporting the status log file to centralized application performance tool **110**.

(23) Centralized application performance tool **110** receives status indicators for one or more of applications **104** running on server **108**A of data center **102**, and may further receive status indicators for other enterprise applications running on servers within any of data centers **102** and/or on mainframe computer **112**. In one specific example, application performance tool **110** may read or otherwise receive status log files including the status indicators and the corresponding metric values from the applications. In some examples, application performance tool **110** may receive the status indicators for the applications via third-party monitoring tool **124** included in enterprise network **100**. According to the disclosed techniques, application performance tool **110** may perform comparisons or other analyses of the status indicators and/or the metric values across multiple instances of the same application in order to verify which instance is the active or primary instance and/or which instances are the healthiest.

(24) In some examples, application performance tool **110** may generate a user interface to display locations (e.g., one of data centers **102** and/or mainframe **112**) and statuses of the plurality of applications running in enterprise network **100**. The user interface may enable a user, e.g., an administrator of enterprise network **100** via console **126**, to view the application statuses from a data center perspective of what applications are running on a certain data center, and/or from an application perspective of where each instance of a given application is running.

(25) In the example of FIG. **1**, enterprise network **100** might include all of the components shown in FIG. **1**. Further, in some examples, an enterprise network may not include third-party monitoring tool **124**. The optional nature of third-party monitoring tool **124** is indicated through the use of a dashed outline and dashed lines connecting third-party monitoring tool **124** to data centers **102** and application performance tool **110**.

(26) Each of the computing systems illustrated in FIG. **1** (e.g., application performance tool **110**, user devices **116**, console **126**, data store **128**, third-party monitoring tool **124**) may represent any suitable computing system, such as one or more server computers, cloud computing systems, mainframes, appliances, desktop computers, laptop computers, mobile devices, and/or any other computing device that may be capable of performing operations in accordance with one or more aspects of the present disclosure. One or more of such devices may perform operations described herein as a result of instructions, stored on a computer-readable storage medium, executing on one or more processors. The instructions may be in the form of software stored on one or more local or remote computer readable storage devices. In other examples, one or more of such computing devices may perform operations using hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at each of such computing devices.

(27) FIG. **2** is a block diagram illustrating an example computing device **208** executing applications **204**A-**204**M (collectively, "applications **204**") having embedded status modules **206**A-**206**M (collectively, "status modules **206**") configured to self-report application statuses, in accordance with one or more aspects of the present disclosure. Computing device **208** may generally correspond any of servers **108** of data center **102**A, other servers within any of data centers **102**B-**102**K, or mainframe computer **112** from FIG. **1**. Accordingly, status modules **206** may perform some or all of the same functions described as being performed by status modules **106** from FIG. **1**.
(28) Computing device **208** may be implemented as any suitable computing system, such as one or more server computers, workstations, mainframes, appliances, cloud computing systems, and/or other computing systems that may be capable of performing operations and/or functions described in accordance with one or more aspects of the present disclosure. In some examples, computing device **208** may comprise a server within a data center, cloud computing system, server farm, and/or server cluster (or portion thereof) that provides services to client devices and other devices or systems. For example, computing device **208** may host or provide access to services provided by one or more applications **204** running on computing device **208**.
(29) Although computing device **208** of FIG. **2** is illustrated as a stand-alone device, in other examples computing device **208** may be implemented in any of a wide variety of ways, and may be implemented using multiple devices and/or systems. In some examples, computing device **208** may be, or may be part of, any component, device, or system that includes a processor or other suitable computing environment for processing information or executing software instructions and that operates in accordance with one or more aspects of the present disclosure. In some examples, computing device **208** may be fully implemented as hardware in one or more devices or logic elements.
(30) In the example of FIG. **2**, computing device **208** may include one or more processors **210**, one or more communication units **212**, one or more input/output devices **214**, and one or more storage devices **216**. Storage devices **216** may include applications **204** having embedded status modules **206**, application logs **220**A-**220**M (collectively, "application logs **220**") corresponding to respective applications **204**, status logs **224**A-**224**M (collectively, "status logs **224**") corresponding to respective applications **204**, and an API **230**. One or more of the devices, modules, storage areas, or other components of computing device **208** may be interconnected to enable inter-component communications (physically, communicatively, and/or operatively). In some examples, such connectivity may be provided by through communication channels, a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data. A power source (not shown) is provide power to one or more components of computing device **208**. In some examples, the power source may receive power from the primary alternative current (AC) power supply in a commercial building or data center, where some or all of an enterprise network may reside. In other examples, the power source may be or may include a battery.
(31) One or more processors **210** of computing device **208** may implement functionality and/or execute instructions associated with computing device **208** associated with one or more modules illustrated herein and/or described below. One or more processors **210** may be, may be part of, and/or may include processing circuitry that performs operations in accordance with one or more aspects of the present disclosure. Examples of processors **210** include microprocessors, application processors, display controllers, auxiliary processors, one or more sensor hubs, and any other hardware configured to function as a processor, a processing unit, or a processing device. Computing device **208** may use one or more processors **210** to perform operations in accordance with one or more aspects of the present disclosure using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at computing device **208**.
(32) One or more communication units **212** of computing device **208** may communicate with devices external to computing device **208** by transmitting and/or receiving data, and may operate,

in some respects, as both an input device and an output device. In some examples, communication units **212** may communicate with other devices over a network. In other examples, communication units **212** may send and/or receive radio signals on a radio network such as a cellular radio network. In other examples, communication units **212** of computing device **208** may transmit and/or receive satellite signals on a satellite network such as a Global Positioning System (GPS) network. Examples of communication units **212** include a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GPS receiver, or any other type of device that can send and/or receive information. Other examples of communication units **245** may include devices capable of communicating over Bluetooth®, GPS, NFC, ZigBee, and cellular networks (e.g., 3G, 4G, 5G), and Wi-Fi® radios found in mobile devices as well as Universal Serial Bus (USB) controllers and the like. Such communications may adhere to, implement, or abide by appropriate protocols, including Transmission Control Protocol/Internet Protocol (TCP/IP), Ethernet, Bluetooth, NFC, or other technologies or protocols.
(33) One or more input/output devices **214** may represent any input or output devices of computing device **208** not otherwise separately described herein. One or more input/output devices **214** may generate, receive, and/or process input from any type of device capable of detecting input from a human or machine. One or more input/output devices **214** may generate, present, and/or process output through any type of device capable of producing output.
(34) One or more storage devices **216** within computing device **208** may store information for processing during operation of computing device **208**. Storage devices **216** may store program instructions and/or data associated with one or more of the modules described in accordance with one or more aspects of this disclosure. One or more processors **210** and one or more storage devices **216** may provide an operating environment or platform for such modules, which may be implemented as software, but may in some examples include any combination of hardware, firmware, and software. One or more processors **210** may execute instructions and one or more storage devices **216** may store instructions and/or data of one or more modules. The combination of processors **210** and storage devices **216** may retrieve, store, and/or execute the instructions and/or data of one or more applications, modules, or software. Processors **210** and/or storage devices **216** may also be operably coupled to one or more other software and/or hardware components, including, but not limited to, one or more of the components of computing device **208** and/or one or more devices or systems illustrated as being connected to computing device **208**.
(35) In some examples, one or more storage devices **216** are temporary memories, meaning that a primary purpose of the one or more storage devices is not long-term storage. Storage devices **216** of computing device **208** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if deactivated. Examples of volatile memories include random access memories (RAM), dynamic random access memories (DRAM), static random access memories (SRAM), and other forms of volatile memories known in the art. Storage devices **216**, in some examples, also include one or more computer-readable storage media. Storage devices **216** may be configured to store larger amounts of information than volatile memory. Storage devices **216** may further be configured for long-term storage of information as non-volatile memory space and retain information after activate/off cycles. Examples of non-volatile memories include magnetic hard disks, optical discs, floppy disks, Flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories.
(36) Each of applications **204** may comprise an instance of a different enterprise application developed by a line of business for execution on a shared enterprise network, e.g., enterprise network **100** from FIG. **1**. When applications **204** are deployed and in use, in response to receipt of a request from a user device via communication units **212**, processors **210** may execute an appropriate one of applications **204**, e.g., application **204**A, to perform the requested operations and/or services. In addition, application **204**A may log events related to the performance of the

operations and services in one or more corresponding application logs **220**A. In some examples, application **204**A may compute certain metrics based on the logged events recorded in application logs **220**A over time. For example, application **104**A may continuously or periodically compute metrics including service call volume, service call speed, and/or data write frequency based on activity in application logs **220**A, CPU throughput of computing device **208**, and/or IO speed of computing device **208**.

(37) One or more of applications **204** may comprise instances of different distributed applications having multiple instances running in geographically-disparate computing devices (e.g., data center servers or mainframe computers) of the enterprise network. In one example, the distributed applications may have active-standby instances in which one of the instances operates as the active instance that performs operations and services on behalf of user devices while the standby instances remain idle or perform administrative operations. In another example, the distributed applications may have active-active instances in which a primary active instance performs operations and services on behalf of user devices while the other active instances perform the same operations or sync with the primary active instance (e.g., to provide redundancy between the application instances).

(38) As discussed above, the available tools for application discovery and monitoring may only be able to discover which applications are running in a network environment, and not where the applications are running (e.g., data center servers and/or mainframe computers) and/or where the active or primary instance of a distributed application is running. To enable better planning and responsiveness in the case of adverse events within large-scale enterprise networks, this disclosure describes use of status modules **206** embedded in applications **204** that comprise lightweight code modules configured to self-report statuses of the respective applications **204**.

(39) According to the disclosed techniques, a given one of status modules **206**, e.g., status module **206**A embedded within application **204**A, is configured to periodically determine an operational status of respective application **204**A based on at least one value of at least one metric of application **204**A for each periodic interval, and self-report a status indicator representative of the operational status of application **204**A at a given time to a centralized application performance tool, e.g., application performance tool **110** from FIG. **1**. In some examples, status module **206**A is configured to record at least the status indicator for application **204**A, and in some cases the value of the metric used to determine the operational status of application **204**A for each periodic interval, into a timestamped entry in a status log **224**A corresponding to application **204**A. Status logs **224** may comprise dedicated status logs corresponding to applications **204**.

(40) Each of status modules **206** for a given platform may be programmed or configured with a predefined list of metrics based on which the status module is able to determine an operational status of applications for the given platform. Upon installation of status modules **206** within applications **204**, each of status modules **206** may be initialized to determine the operational status of their specific application based on one or more specific metrics selected from the predefined list of metrics. An enterprise data store, e.g., data store **128** of FIG. **1**, may maintain which metrics of applications **204** are indicative of an operational status (e.g., active/inactive, healthy/unhealthy, or a composite health score) of each respective application and, in some cases, thresholds and/or ranges of metric values associated with certain operational statuses for each respective application. For example, status module **206**A may receive a message from the application performance tool via communication units **212** identifying the metric or metrics that are indicative of the operational status of the specific application **204**A. As one example, the message may include a set of binary flags corresponding to the predefined list of metrics in which the flags corresponding to the selected metric or metrics are set. In some examples, the message may also identify thresholds and/or ranges of metric values associated with certain operational statuses. Status module **206**A may then select the metric or metrics from the predefined list of metrics based on the initialization message.

(41) FIG. **3** is a block diagram illustrating an example of status module **206**A embedded in application **204**A from FIG. **2** in further detail. In the illustrated example of FIG. **3**, status module **206**A includes an application status engine **336**, a composite score engine **338**, an optional monitoring unit **332**, and an optional reporting unit **334**.
(42) Upon initialization, status module **206**A may periodically determine an operational status of application **204**A according to a periodic interval of, e.g., 30 seconds or 1 minute. To determine the operational status, application status engine **336** and/or composite score engine **338** may analyze the selected metric or metrics as recorded in application logs **220**A. In examples where application **204**A may not already be configured to compute the selected metric or metrics for use by status module **206**A, monitoring unit **332** of status module **206**A may periodically or continually monitor or compute the selected metric or metrics of application **204**A. For example, monitoring unit **332** may be configured to monitor a number of log entries within certain application logs **220**A to determine a service call volume metric, monitor a speed at which log entries are being added to certain application logs **220**A to determine a service call speed metric, or monitor a frequency with which certain application logs **220**A are written to a central database, e.g., enterprise data store **128** from FIG. **1**, to determine a data write frequency metric.
(43) Application status engine **336** of status module **206**A may be configured to determine an active status of application **204**A and/or a health status of application **204**A. In some examples, when determining the operational status, in addition to the values of the selected metric or metrics being indicative of the respective statuses of application **204**A, application status engine **336** may also take into account whether the selected metric was computed during an operational period of application **204**A. In general, application **204**A may have certain usage patterns (e.g., scheduled or typical operational periods) that are known to application **204**A. It may be beneficial for application status engine **336** to recognize when application **204**A is in a non-operational period to avoid incorrectly identifying application **204**A as an inactive instance.
(44) In one example, application status engine **336** is configured to determine whether application **204**A is an active instance of a distributed application within the enterprise network based on the value of the selected metric for a given periodic interval meeting a predefined threshold value for the metric, and based on the given periodic interval occurring during a scheduled or typical operational period of application **204**A. In some examples, the selected metric that is indicative of the activity or liveness of application **204**A may be a service call volume metric or a data write frequency metric. As one specific example, application **204**A may comprise an instance of a distributed transaction search application that records service calls performed in response to search requests in one of application logs **220**A. In this example, a service call volume metric, e.g., determined based on a number of log entries within one of application logs **220**A, is indicative of whether application **204**A is an active instance of the distributed transaction search application running in the enterprise network. Application status engine **336** may compare a value of the service call volume metric for application **204**A to the predefined threshold value of the service call volume metric that is indicative of an active instance. If the value of the service call volume metric was computed during the operational period of application **204**A and the value of the service call volume metric meets or exceeds the predefined threshold value, application status engine **336** determines that application **204**A is an active instance of the distributed application.
(45) In another example, application status engine **336** may be configured to determine whether the performance of application **204**A is healthy based on the value of the metric for the given periodic interval being within a predefined value range for the metric, and based on the given periodic interval occurring during the scheduled or typical operational period of application **104**A. In some examples, the selected metric that is indicative of the performance health of application **204**A may be a service call speed metric, a CPU throughput metric, and/or an IO speed metric. Application status engine **336** may identify a healthy performance of application **204**A based on a value of the selected metric being within a predefined value range because, for example, both a low service call

speed and a very high service call speed may be indicative of an anomalous, and thus unhealthy, performance of application **104**A.

(46) Composite score engine **338** of status module **206**A may be configured to compute a composite health score for application **204**A that takes both activity or liveness and performance health into account. For example, composite score engine **338** may be configured to compute a composite health score for application **204**A based on two or more values of two or more metrics, and based on the given periodic interval occurring during a scheduled or typical operational period of application **204**A. In one example, composite score engine **338** may analyze both a service call volume metric and a service call speed metric of application **104**A to compute a numerical value that is indicative of both liveness and performance health of application **204**A.

(47) Upon determining the operational status of application **204**A for a given periodic interval, application status engine **336** and/or composite score engine **338** may then record a status indicator (e.g., a one-bit flag, a multi-bit indicator, or a binary representation of a numerical value) of application **204**A in a timestamped entry in status log **224**A. In addition, application status engine **336**, composite score engine **338**, and/or monitoring unit **332** may record the values of the selected metric or metrics used to determine the operational status of application **204**A for each periodic interval within the timestamped entry in status log **224**A.

(48) Status module **206**A may then self-report an identifier of computing device **208** on which application **204**A is running and at least a status indicator representative of the operational status of application **204**A at a given time that may be periodic (e.g., every 15 minutes or every hour), or may be upon request (e.g., on demand) from the application performance tool and/or a third-party monitoring tool within the enterprise network. The identifier of computing device **208** may include a location of computing device **208** within the enterprise network, e.g., a URL from a domain name system, a data center name, or a mainframe identifier.

(49) Status module **206**A may self-report a single status indicator of application **204**A representative of one of: activity or liveness, performance health, or composite health. In other examples, status module **206**A may self-report two or more status indicators. In some examples, the reported status indicator may be the most recent status indicator recorded in status log **224**A. In other examples, the reported status indicator may include all status indicators and/or all metric values recorded in status log **224**A at the given time. In some examples, status log **224**A may be limited to hold a relatively small volume of entries and may be configured to overwrite the oldest entries. In this way, even if status module **206**A self-reports relatively infrequently (e.g., once per day or once per week), status log **224**A will be maintained at a manageable size.

(50) In some examples, status module **206**A may perform passive self-reporting in that status module **206**A provides read access to the status indicator and/or metric values recorded in status log **224**A via API **230** or another interface of computing device **208**. In other examples, reporting unit **334** of status module **206**A may provide active self-reporting by generating and transmitting a message including the status indicator or writing the status indicator to the central database, e.g., enterprise data store **128** from FIG. **1**. For example, reporting unit **334** may generate a packet having a payload of all timestamped entries in status log **224**A and having a header including the identifier of computing device **208** on which application **204**A is running. Reporting unit **334** may transmit the status indicator, metric values, or the generated packet in a dedicated status reporting message or added onto an existing reporting message between application **204**A and the application performance tool. As another example, reporting unit **334** may store the status indicator and/or metric values in the central database of the enterprise network that is accessible by the application performance tool.

(51) Modules illustrated in FIGS. **2** and **3** (e.g., applications **204**, status modules **206**, application status engine **336**, composite score engine **338**, monitoring unit **332**, and reporting unit **334**) and/or illustrated or described elsewhere in this disclosure may perform operations described using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or

executing at one or more computing devices. For example, a computing device may execute one or more of such modules with multiple processors or multiple devices. A computing device may execute one or more of such modules as a virtual machine executing on underlying hardware. One or more of such modules may execute as one or more services of an operating system or computing platform. One or more of such modules may execute as one or more executable programs at an application layer of a computing platform. In other examples, functionality provided by a module could be implemented by a dedicated hardware device.

(52) Although certain modules, data stores, components, programs, executables, data items, functional units, and/or other items included within one or more storage devices may be illustrated separately, one or more of such items could be combined and operate as a single module, component, program, executable, data item, or functional unit. For example, one or more modules or data stores may be combined or partially combined so that they operate or provide functionality as a single module. Further, one or more modules may interact with and/or operate in conjunction with one another so that, for example, one module acts as a service or an extension of another module. Also, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may include multiple components, sub-components, modules, sub-modules, data stores, and/or other components or modules or data stores not illustrated.

(53) Further, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may be implemented in various ways. For example, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may be implemented as a downloadable or pre-installed application or "app." In other examples, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may be implemented as part of an operating system executed on a computing device.

(54) FIG. **4** is a block diagram illustrating an example computing system **400** executing an application performance tool **410** configured to process application statuses received from a plurality of applications, in accordance with one or more aspects of the present disclosure. Computing system **400** may generally correspond to a device that includes and/or implements aspects of the functionality of application performance tool **110** illustrated in FIG. **1**. Accordingly, computing system **400** executing application performance tool **410** may perform some or all of the same functions described in connection with FIG. **1** as being performed by application performance tool **110**.

(55) Computing system **400** may be implemented as any suitable computing system, such as one or more server computers, workstations, mainframes, appliances, cloud computing systems, and/or other computing systems that may be capable of performing operations and/or functions described in accordance with one or more aspects of the present disclosure. In some examples, computing system **400** represents a cloud computing system, server farm, and/or server cluster (or portion thereof) that provides services to client devices and other devices or systems.

(56) Although computing system **400** of FIG. **4** is illustrated as a stand-alone device, in other examples computing system **400** may be implemented in any of a wide variety of ways, and may be implemented using multiple devices and/or systems. In some examples, computing system **400** may be, or may be part of, any component, device, or system that includes a processor or other suitable computing environment for processing information or executing software instructions and that operates in accordance with one or more aspects of the present disclosure. In some examples, computing system **400** may be fully implemented as hardware in one or more devices or logic elements.

(57) In the example of FIG. **4**, computing system **400** may include one or more processors **402**, one or more communication units **404**, one or more input/output devices **406**, and one or more storage devices **408**. Storage devices **408** may include application performance tool **410** including a status

comparison engine **432**, a status module configuration unit **434**, an application status engine **436**, a composite score engine **438**, a user interface module **440**, an application metric store **412**, and one or more application information records **414**. One or more of the devices, modules, storage areas, or other components of computing system **400** may be interconnected to enable inter-component communications (physically, communicatively, and/or operatively). In some examples, such connectivity may be provided by through communication channels, a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data. A power source (not shown) is provide power to one or more components of computing system **400**. In some examples, the power source may receive power from the primary alternative current (AC) power supply in a commercial building or data center, where some or all of an enterprise network may reside. In other examples, the power source may be or may include a battery.

(58) One or more processors **402** of computing system **400** may implement functionality and/or execute instructions associated with computing system **400** associated with one or more modules illustrated herein and/or described below. One or more processors **402** may be, may be part of, and/or may include processing circuitry that performs operations in accordance with one or more aspects of the present disclosure. Examples of processors **402** include microprocessors, application processors, display controllers, auxiliary processors, one or more sensor hubs, and any other hardware configured to function as a processor, a processing unit, or a processing device. Computing system **400** may use one or more processors **402** to perform operations in accordance with one or more aspects of the present disclosure using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at computing system **400**.

(59) One or more communication units **404** of computing system **400** may communicate with devices external to computing system **400** by transmitting and/or receiving data, and may operate, in some respects, as both an input device and an output device. In some examples, communication units **404** may communicate with other devices over a network. In other examples, communication units **404** may send and/or receive radio signals on a radio network such as a cellular radio network. In other examples, communication units **404** of computing system **400** may transmit and/or receive satellite signals on a satellite network such as a GPS network. Examples of communication units **404** include a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GPS receiver, or any other type of device that can send and/or receive information. Other examples of communication units **404** may include devices capable of communicating over Bluetooth®, GPS, NFC, ZigBee, and cellular networks (e.g., 3G, 4G, 5G), and Wi-Fi® radios found in mobile devices as well as USB controllers and the like. Such communications may adhere to, implement, or abide by appropriate protocols, including TCP/IP, Ethernet, Bluetooth, NFC, or other technologies or protocols.

(60) One or more input/output devices **406** may represent any input or output devices of computing system **400** not otherwise separately described herein. One or more input/output devices **406** may generate, receive, and/or process input from any type of device capable of detecting input from a human or machine. One or more input/output devices **406** may generate, present, and/or process output through any type of device capable of producing output.

(61) One or more storage devices **408** within computing system **400** may store information for processing during operation of computing system **400**. Storage devices **408** may store program instructions and/or data associated with one or more of the modules described in accordance with one or more aspects of this disclosure. One or more processors **402** and one or more storage devices **408** may provide an operating environment or platform for such modules, which may be implemented as software, but may in some examples include any combination of hardware, firmware, and software. One or more processors **402** may execute instructions and one or more storage devices **408** may store instructions and/or data of one or more modules. The combination of processors **402** and storage devices **408** may retrieve, store, and/or execute the instructions and/or

data of one or more applications, modules, or software. Processors **402** and/or storage devices **408** may also be operably coupled to one or more other software and/or hardware components, including, but not limited to, one or more of the components of computing system **400** and/or one or more devices or systems illustrated as being connected to computing system **400**.

(62) In some examples, one or more storage devices **408** are temporary memories, meaning that a primary purpose of the one or more storage devices is not long-term storage. Storage devices **408** of computing system **400** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if deactivated. Examples of volatile memories include RAM, DRAM, SRAM, and other forms of volatile memories known in the art. Storage devices **408**, in some examples, also include one or more computer-readable storage media. Storage devices **408** may be configured to store larger amounts of information than volatile memory. Storage devices **408** may further be configured for long-term storage of information as non-volatile memory space and retain information after activate/off cycles. Examples of non-volatile memories include magnetic hard disks, optical discs, floppy disks, Flash memories, or forms of EPROM or EEPROM memories.

(63) According to the disclosed techniques, application performance tool **410** of computing system **400** is configured to receive or retrieve status indicators for one or more of applications running on geographically-disparate data center servers and/or mainframe computers within the enterprise network, and analyze the status indicators to identify, for a particular application, a location of a primary instance of the particular application within the enterprise network. In some examples, application performance tool **410** may verify which instance of the particular application is the active or primary instance and/or which instances of the particular application are the healthiest.

(64) As described above, the status modules embedded in each of the applications may be programmed or configured with a predefined list of metrics based on which the status module is able to determine an operational status of applications for the given platform. To initialize the status modules, status module configuration unit **434** of application performance tool **410** may send a message to each of the status modules identifying the at least one metric from the predefined list of metrics that is indicative of the operational status of the respective application. As one example, the message may include a set of binary flags corresponding to the predefined list of metrics in which the flags corresponding to the selected metric or metrics are set. In some examples, the message may also identify thresholds and/or ranges of metric values associated with certain operational statuses.

(65) In some examples, status module configuration unit **434** may perform functions relating to maintaining, updating, and interacting with an enterprise data store, e.g., data store **128** from FIG. **1**. Status module configuration unit **434** may maintain application information records **414** within the enterprise data store, and may update application information records **414** and/or the enterprise data store in response to input. For instance, status module configuration unit **434** may receive input from a computing device associated with one or more lines of business. Status module configuration unit **434** may determine that the input corresponds to information about one or more enterprise applications administered, developed, or updated by such lines of business. Status module configuration unit **434** may also receive input from a console, e.g., console **126** from FIG. **1**, or from another source. Status module configuration unit **434** may update application information records **414** and/or the enterprise data store as enterprise applications are modified, further developed, or otherwise evolved.

(66) Application information records **414** may represent one or more files, records, or other storage units that include information about any changes, modifications, or updates that are made enterprise applications. Application information records **414** may be primarily maintained by status module configuration unit **434** so that each of application information records **414** includes relatively up-to-date information about its corresponding application; such information may include which metrics of the enterprise applications are indicative of an operational status (e.g.,

active/inactive, healthy/unhealthy, or a composite health score) of each respective application and, in some cases, thresholds and/or ranges of metric values associated with certain operational statuses for each respective application.

(67) After the status modules are initiated, application performance tool **410** may receive the status indicators, and in some cases the corresponding metric values, for the one or more applications running in the enterprise network via communication unit **404** of computing system **400**. In addition, along with each status indicator for a respective application, application performance tool **410** may also receive an identifier of the computing device on which the respective application is running within the enterprise network. The identifier may include a location of the respective computing device within the enterprise network, e.g., a URL from a domain name system, a data center name, or a mainframe identifier.

(68) Application performance tool **410** may receive the status indicators in messages received from the one or more applications. In one specific example, application performance tool **410** may read or otherwise receive status log files including the status indicators and corresponding metric values from one or more of the applications. In some examples, application performance tool **410** may receive the status indicators for the applications via a third-party monitoring tool, e.g., third-party monitoring tool **124**, included in the enterprise network. Upon receipt of the status indicators according to the above examples, application performance tool **410** may store the received status indicators in application metric store **412**, which may comprise a portion of the enterprise data store. In further examples, the one or more applications may write the status indicators and corresponding metric values directly to application metric store **412** within the enterprise data store, such that application performance tool **410** may simply read the status indicators from application metric store **412**.

(69) Status comparison engine **432** of application performance tool **410** may perform comparisons or other analyses of the status indicators and/or the metric values across multiple instances of the same application in order to verify which instance is the active or primary instance and/or which instances are the healthiest. For example, status comparison engine **432** analyzes the status indicators received from the status modules embedded in instances of a particular application running on a plurality of computing devices within the enterprise network to identify the location of the primary instance of the particular application. The location of the primary instance of the particular application may be determined based on the identifier of the computing device on which the primary instance of the particular application is running, where the computing device comprises one of a particular data center server or a particular mainframe computer within the enterprise network.

(70) As one specific example, application performance tool **410** may receive a first status indicator for a first application and a first metric value used to determine a first operational status of the first application, where the first application is a first instance of the particular application running on a first computing device. Application performance tool **410** may also receive a second status indicator for a second application and a second metric value used to determine a second operational status of the second application, where the second application is a second instance of the particular application running on a second computing device. In a scenario where both the first application and the second application are identified as being active instances of the particular application, status comparison engine **432** may determine that the first application is the primary active instance of the particular application based on performing a comparison between the first metric value of the first application and the second metric value of the second application. For example, if both the first application and the second application had metric values that meet the predefined threshold for the selected metric of the particular application, both applications may be indicated as being active instances, but the primary active instance may be identified by status comparison engine **432** as have a higher metric value than other non-primary active instances. In addition, status comparison engine **432** may identify the location of the primary instance of the particular application based on

the identifier of the first computing device on which the first application is running.
(71) In some examples, application performance tool **410** includes an application status engine **436** and a composite score engine **438** that may operate substantially similar to application status engine **336** and composite score engine **338** of status module **206**A form FIG. **3**. For example, application status engine **436** may be configured to determine an active status of the particular application and/or a health status of the particular application based on the metric values received from the particular application. Composite score engine **438** may be configured to compute a composite health score for the particular application that takes both activity or liveness and performance health into account.
(72) As one specific example, application performance tool **410** may receive the status indicator for the particular application and the value of the metric of the particular application used to determine a first operational status of the particular application. Application status engine **436** of application performance tool **410** may determine a second operational status of the particular application based on the received value of the metric of the particular application. Status comparison engine **432** may then verify whether the received status indicator for the particular application is accurate based on performing a comparison between the first operational status and the second operational status of the particular application. In scenarios where the first operational status and the second operational status of the particular application are different, e.g., the status module embedded in the particular application determined that the particular application is an inactive instance and the application status engine **436** of application performance tool **410** determined that the particular application is an active instance, the anomaly may indicate a network issue or configuration issue with the status module of the particular application. In either case, such an anomaly may trigger a notification to an administrator of the enterprise network to investigate the cause of the anomaly.
(73) UI module **440** of application performance tool **110** may generate data representative of a user interface to display on a user interface device, e.g., console **126** of FIG. **1**, locations (e.g., data center servers and/or mainframe computers) and statuses of the plurality of applications running in the enterprise network. UI module **440** may present to a user e.g., an administrator of the enterprise network via the user interface device, a view of the application statuses from a data center perspective of what applications are running on a certain data center, and/or from an application perspective of where each instance of a given application is running.
(74) In one example, UI module **440** may generate a graphical user interface (GUI) comprising a searchable and sortable table with rows of applications and columns of locations (e.g., data center servers and mainframes). The table may include color-coded cells to indicate the locations of active instances of the applications. In other examples, the table may include words or symbols in each of the cells to indicate which application instances are active, inactive, or have no presence in each location. In some examples, the table may further include an indication of the health status and/or a numerical value of the composite health score in the corresponding cells for each of the one or more applications. The table may be searchable and sortable by application or by location.
(75) Modules illustrated in FIG. **4** (e.g., status comparison engine **432**, status module configuration unit **434**, application status engine **436**, composite score engine **438**, UI module **440**) and/or illustrated or described elsewhere in this disclosure may perform operations described using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at one or more computing devices. For example, a computing device may execute one or more of such modules with multiple processors or multiple devices. A computing device may execute one or more of such modules as a virtual machine executing on underlying hardware. One or more of such modules may execute as one or more services of an operating system or computing platform. One or more of such modules may execute as one or more executable programs at an application layer of a computing platform. In other examples, functionality provided by a module could be implemented by a dedicated hardware device.
(76) Although certain modules, data stores, components, programs, executables, data items,

functional units, and/or other items included within one or more storage devices may be illustrated separately, one or more of such items could be combined and operate as a single module, component, program, executable, data item, or functional unit. For example, one or more modules or data stores may be combined or partially combined so that they operate or provide functionality as a single module. Further, one or more modules may interact with and/or operate in conjunction with one another so that, for example, one module acts as a service or an extension of another module. Also, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may include multiple components, sub-components, modules, sub-modules, data stores, and/or other components or modules or data stores not illustrated.

(77) Further, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may be implemented in various ways. For example, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may be implemented as a downloadable or pre-installed application or “app.” In other examples, each module, data store, component, program, executable, data item, functional unit, or other item illustrated within a storage device may be implemented as part of an operating system executed on a computing device.

(78) FIG. **5** is a flow diagram illustrating operations performed by an example computing executing one or more applications and embedded status modules, in accordance with one or more aspects of the present disclosure. The operations of FIG. **5** are described within the context of computing device **208** and status module **206**A embedded within application **204**A from FIGS. **2** and **3**. In other examples, operations described in FIG. **5** may be performed by servers **108** from FIG. **1**, or one or more other components, modules, systems, or devices. Further, in other examples, operations described in connection with FIG. **5** may be merged, performed in a difference sequence, or omitted.

(79) Status modules **206** may be configured or programmed with a predefined list of metrics based on which the status modules are able to determine an operational status of applications. For a first status module **206**A, computing device **208** may receive a message from an application performance tool, e.g., application performance tool **110** from FIG. **1**, identifying at least one metric from the list of predefined metrics that is indicative of the operational status of respective application **204**A (**505**). In some examples, the message may also identify thresholds and/or ranges of metric values associated with certain operational statuses. In response to the message, status module **206**A selects the at least one metric identified in the message for use to determine the operational status of first application **204**A (**510**).

(80) Upon initialization, status module **206**A periodically determines an operational status of first application **204**A based on at least one value of the selected metric of first application **204**A for each periodic interval (**515**). Status module **206**A then reports an identifier of computing device **208** on which first application **204**A is running and at least a status indicator representative of the operational status of first application **204**A at a given time to the application performance tool within the enterprise network (**520**).

(81) In one specific example, for a given periodic interval, status module **206**A determines whether first application **204**A comprising an active instance based on the value of the selected metric meeting a predefined threshold value for the selected metric and whether the given periodic interval occurs during an operational period of the first application. Upon determining the operational status of first application **206**A for each periodic interval, status module **206**A may record the status indicator for first application **204**A and the value of the selected metric used to determine the operational status of first application **204**A into a timestamped entry in a status log **224**A corresponding to first application **204**A. Status module **206**A may then generating a packet having a payload of all timestamped entries in status log **224**A at the given time and having a header including the identifier of computing device **208** on which first application **204**A is running, and

report the packet to the application performance tool. To report the packet, status module **206**A may transmit the packet in a message to the application performance tool, write or store the packet in a central database of the enterprise network, e.g., data store **128** of FIG. **1**, that is accessible by the application performance tool, or provide read access to status log **224**A via an interface of first application **204**A for at least one of the application performance tool or a third-party monitoring tool within the enterprise network.

(82) FIG. **6** is a flow diagram illustrating operations performed by an example computing system executing an application performance tool, in accordance with one or more aspects of the present disclosure. The operations of FIG. **6** are described within the context of computing system **400** executing application performance tool **410** from FIG. **4**. In other examples, operations described in FIG. **6** may be performed by application performance tool **110** from FIG. **1**, or one or more other components, modules, systems, or devices. Further, in other examples, operations described in connection with FIG. **6** may be merged, performed in a difference sequence, or omitted.

(83) In accordance with to the disclosed techniques, status modules embedded in applications may be configured or programmed with a predefined list of metrics based on which the status modules are able to determine an operational status of applications. To initialize the status modules, application performance tool **410** may send a message to each status module embedded within a respective application (e.g., status modules **106** of applications **104** from FIG. **1**) identifying at least one metric from the list of predefined metrics that is indicative of the operational status of the respective application (**605**). In some examples, the message may also identify thresholds and/or ranges of metric values associated with certain operational statuses.

(84) Upon initialization, application performance tool **410** receives status indicators for applications running on a plurality of computing devices (e.g., data center severs and/or mainframe computers) within an enterprise network from status modules embedded in the applications (**610**). Application performance tool **410** is configured to analyze the received status indicators to identify, for a particular application at a given time, a location of a primary instance of the particular application within the enterprise network (**615**). The location of the primary instance of the particular application may be determined based on an identifier of the computing device on which the primary instance of the particular application is running that is reported with the status indicators for the particular application. The identifier of the computing device may include, e.g., a URL from a domain name system, a data center name, or a mainframe identifier. Application performance tool **410** may then generate data representative of a user interface for display on a user interface device, e.g., console **126** from FIG. **1**, to present locations and operational statuses of the applications running on the plurality of computing devices within the enterprise network (**620**).

(85) In one example, application performance tool **410** receive a first status indicator for a first application and a first metric value used to determine a first operational status of the first application, the first application comprising a first instance of the particular application running on a first computing device. Application performance tool **410** also receives a second status indicator for a second application and a second metric value used to determine a second operational status of the second application, the second application comprising a second instance of the particular application running on a second computing device. Application performance tool **410** then determines that the first application is the primary instance of the particular application based on performing a comparison between the first metric value of the first application and the second metric value of the second application, and identifies the location of the primary instance of the particular application based on an identifier of the first computing device on which the first application is running.

(86) In another example, application performance tool **410** receives a status indicator for a respective application and a value of the metric of the respective application used to determine a first operational status of the respective application from the status module embedded within the respective application. Application performance tool **410** then determines a second operational

status of the respective application based on the received value of the metric of the respective application. Application performance tool **410** verifies whether the received status indicator for the respective application is accurate based on performing a comparison between the first operational status and the second operational status of the respective application. In scenarios where the first operational status and the second operational status do not match, application performance tool **410** may trigger a notification to an administrator of the enterprise network to investigate the cause of the operational status anomaly.

(87) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media, or communication media including any medium that facilitates transfer of computer program from one place to another, e.g., according to a communication protocol. In this manner, computer-readable storage media generally may correspond to (1) tangible computer-readable storage media which is non-transitory or (2) a communication media such as signal or carrier wave. Data storage media may be any available media that can be accessed by one or more computers or one or more processing circuits to receive instructs, code and/or data structures for implementation of the techniques described in this disclosure. A computer program product may include a computer-readable medium.

(88) By way of example and not limitation, such computer-readable storage media may include RAM, ROM, EEPROM, CD-ROM, or other optical disk storage, magnetic disk storage, or other magnetic storage devices, flash memory, cache memory, or any other medium that can be used to store desired program code in the form of instructions or store data structures and that can be access by a computer. Also, any connection is a properly termed a computer-readable medium. For example, if instructions are transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or other wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or other wireless technologies such as infrared, radio, and microwave are included in the definition of medium. It should be understood, however, that computer-readable storage media and data storage media do not include connections, carrier waves, signals, or other transient media, but are directed to non-transient, tangible storage media. Disk and disc, as used herein, includes compact disk (CD), laser disc, optical disc, digital versatile disc (DVD), and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should be included within the scope of computer-readable media.

(89) Functionality described in this disclosure may be performed by fixed function and/or programmable processing circuitry. For instance, instructions may be executed by fixed function and/or programmable processing circuitry. Such processing circuitry may include one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor”, as used herein may refer to any of the foregoing structure of any other structure suitable for implementation of the techniques described herein. In addition, in some aspects, the functionality described herein may be provided within dedicated hardware and/or software modules. Also, the techniques could be fully implemented in one or more circuits or logic elements. Processing circuits may be coupled to other components in various ways. For example, a processing circuit may be coupled to other components via an internal device interconnect, a wired or wireless network connection, or another communication medium.

(90) The techniques of this disclosure may be implemented in a wide variety of devices or

apparatuses, an integrated circuit (IC) or a set of ICs (e.g., a chip set). Various components, modules, software systems, or units are described in this disclosure to emphasize functional aspects of devices configured to perform the disclosed techniques, but do not necessarily require realization by different hardware units. Rather, as described above, various units may be combined in a hardware unit or provided by a collection of interoperative hardware units, including one or more processors as described above, in conjunction with suitable software and/or firmware.

Claims

1. A computing device within an enterprise network, the computing device comprising: a storage device configured to store one or more applications and one or more status modules embedded within the one or more applications; and processing circuitry having access to the storage device and configured to execute instructions associated with the one or more status modules embedded within the one or more applications, wherein to execute instructions associated with a first status module embedded within a first application, the processing circuitry is further configured to: receive, by the first status module from an application performance tool within the enterprise network, a message identifying at least one metric from a predefined list of metrics programmed on each of the one or more status modules as indicative of an operational status of the first application, wherein the operational status includes one of an active status, a health status, or a composite health score, wherein the message includes a set of binary flags that correspond to the predefined list of metrics, and wherein at least one flag corresponding to the at least one metric is set to identify the at least one metric as indicative of the operational status of the first application; select, by the first status module, the at least one metric for the first application from the predefined list of metrics based on the message received from the application performance tool; periodically determine, by the first status module, the operational status of the first application based on at least one value of the at least one metric of the first application for each periodic interval; and report, by the first status module, an identifier of the computing device on which the first application is running and at least a status indicator representative of the operational status of the first application at a given time to the application performance tool within the enterprise network.
2. The computing device of claim 1, wherein the storage device stores one or more status logs corresponding to the one or more applications, and wherein to execute instructions associated with the first status module, the processing circuitry is further configured to, upon determining the operational status of the first application for each periodic interval, record at least the status indicator for the first application into a timestamped entry in a first status log of the one more status logs corresponding to the first application.
3. The computing device of claim 2, wherein to execute instructions associated with the first status module, the processing circuitry is further configured to record the at least one value of the at least one metric used to determine the operational status of the first application for each periodic interval within the timestamped entry along with the status indicator for the first application.
4. The computing device of claim 2, wherein to report at least the status indicator for the first application at the given time, the processing circuitry is further configured to: generate a packet having a payload of one or more timestamped entries in the first status log at the given time and having a header including the identifier of the computing device on which the first application is running; and report the packet to the application performance tool.
5. The computing device of claim 1, wherein to periodically determine the operational status of the first application, the processing circuitry is further configured to one of periodically or continually monitor the at least one metric of the first application.
6. The computing device of claim 1, wherein the operational status comprises the active status, and wherein to determine the operational status of the first application for a given periodic interval, the processing circuitry is further configured to determine whether the first application is an active

instance based on the at least one value of the at least one metric meeting a predefined threshold value for the at least one metric and whether the given periodic interval occurs during an operational period of the first application.
7. The computing device of claim 1, wherein the operational status comprises the health status, and wherein to determine the operational status of the first application for a given periodic interval, the processing circuitry is further configured to determine performance health of the first application based on the at least one value of the at least one metric being within a predefined value range for the at least one metric and whether the given periodic interval occurs during an operational period of the first application.
8. The computing device of claim 1, wherein the operational status comprises the composite health score, and wherein to determine the operational status of the first application for a given periodic interval, the processing circuitry is further configured to determine the composite health score for the first application based on two or more values of two or more metrics and whether the given periodic interval occurs during an operational period of the first application.
9. The computing device of claim 1, wherein to report the status indicator for the first application at the given time, the processing circuitry is further configured to, periodically or in response to a request, at least one of: transmit the status indicator in a message to the application performance tool; store the status indicator in a central database of the enterprise network that is accessible by the application performance tool; or provide read access to the status indicator via an interface of the first application for at least one of the application performance tool or a third-party monitoring tool within the enterprise network.
10. A method comprising: receiving, by a first status module embedded within a first application of one or more status modules embedded within one or more applications running on a computing device within an enterprise network and from an application performance tool, a message identifying at least one metric from a predefined list of metrics programmed on each of the one or more status modules as indicative of an operational status of the first application, wherein the operational status includes one of an active status, a health status, or a composite health score, wherein the message includes a set of binary flags that correspond to the predefined list of metrics, and wherein at least one flag corresponding to the at least one metric is set to identify the at least one metric as indicative of the operational status of the first application; selecting, by the first status module, the at least one metric for the first application from the predefined list of metrics based on the message received from the application performance tool; periodically determining, by the first status module, the operational status of the first application based on at least one value of the at least one metric of the first application for each periodic interval; and reporting, by the first status module, an identifier of the computing device on which the first application is running and at least a status indicator representative of the operational status of the first application at a given time to the application performance tool within the enterprise network.
11. The method of claim 10, further comprising, upon determining the operational status of the first application for each periodic interval, recording at least the status indicator for the first application into a timestamped entry in a first status log corresponding to the first application.
12. The method of claim 11, further comprising recording the at least one value of the at least one metric used to determine the operational status of the first application for each periodic interval within the timestamped entry along with the status indicator for the first application.
13. The method of claim 11, wherein reporting at least the status indicator for the first application at the given time further comprises: generating a packet having a payload of one or more timestamped entries in the first status log at the given time and having a header including the identifier of the computing device on which the first application is running; and reporting the packet to the application performance tool.
14. The method of claim 10, wherein the operational status comprises the active status, and wherein determining the operational status of the first application for a given periodic interval comprises

determining whether the first application is an active instance based on the value of the metric meeting a predefined threshold value for the metric and whether the given periodic interval occurs during an operational period of the first application.

15. The method of claim 10, wherein reporting the status indicator for the first application at the given time comprises, periodically or in response to a request, at least one of: transmitting the status indicator in a message to the application performance tool; storing the status indicator in a central database of the enterprise network that is accessible by the application performance tool; or providing read access to the status indicator via an interface of the first application for at least one of the application performance tool or a third-party monitoring tool within the enterprise network.

16. A network system comprising: an application performance tool; and a plurality of computing devices, each computing device configured to execute one or more status modules embedded within one or more instances of a distributed application, wherein each status module embedded within a respective instance of the distributed application running on a respective computing device is configured to: receive, from the application performance tool, a message identifying at least one metric from a predefined list of metrics programmed on each of the one or more status modules as indicative of an operational status of the respective instance of the distributed application, wherein the operational status includes one of an active status, a health status, or a composite health score, wherein the message includes a set of binary flags that correspond to the predefined list of metrics, and wherein at least one flag corresponding to the at least one metric is set to identify the at least one metric as indicative of the operational status of the first application, periodically determine an operational status of the respective instance of the distributed application based on at least one value of the at least one metric of the respective instance of the distributed application for each periodic interval, and report an identifier of the respective computing device on which the respective instance of the distributed application is running and at least a status indicator representative of the operational status of the respective instance of the distributed application at a given time to the application performance tool, wherein the application performance tool is configured to analyze status indicators received from the status modules running on the plurality of computing devices to identify, for the distributed application at the given time, a location of a primary instance of the distributed application within the network system, wherein the primary instance of the distributed application has a higher value of the at least one metric than other non-primary instances of the distributed application.

17. The network system of claim 16, wherein the application performance tool is configured to send the message to each of the status modules identifying the at least one metric from the predefined list of metrics that is indicative of the operational status of the respective instance of the distributed application.

18. The network system of claim 16, wherein the location of the primary instance of the distributed application comprises a computing device of the plurality of computing devices within the network system on which the primary instance of the distributed application is running, wherein the computing device comprises one of a data center server or a mainframe computer within the network system.

19. The network system of claim 16, wherein to identify the location of the primary instance of the distributed application at the given time, the application performance tool is configured to: receive a first status indicator for a first application and a first metric value used to determine a first operational status of the first application, the first application comprising a first instance of the distributed application running on a first computing device; receive a second status indicator for a second application and a second metric value used to determine a second operational status of the second application, the second application comprising a second instance of the distributed application running on a second computing device; determine that the first application is the primary instance of the distributed application based on performing a comparison between the first metric value of the first application and the second metric value of the second application, wherein

the first application is the primary instance based on the first metric value of the first application having a higher value than the second metric value of the second application; and identifying the location of the primary instance of the distributed application based on an identifier of the first computing device on which the first application is running.
20. The network system of claim 16, wherein the operational status of the respective instance of the distributed application comprises a first operational status of the respective instance of the distributed application, and wherein the application performance tool is configured to: receive the status indicator for the respective instance of the distributed application and the at least one value of the at least one metric of the respective instance of the distributed application used to determine the first operational status of the respective instance of the distributed application; determine a second operational status of the respective instance of the distributed application based on the received at least one value of the at least one metric of the respective instance of the distributed application; and verify whether the received status indicator for the respective instance of the distributed application is accurate based on performing a comparison between the first operational status and the second operational status of the respective instance of the distributed application.
21. The network system of claim 16, wherein the application performance tool is configured to generate data representative of a user interface for display on a user interface device to present locations and operational statuses of the one or more instances of the distributed application running on the plurality of computing devices within the network system.